

Harsha Halgamuwe Hewage

Forecasting and Supply Chain Analytics

Data Lab for Social Good Research Group, Cardiff Business School, UK

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I am a **supply chain analytics PhD researcher (expected June 2026)** working on forecasting, inventory decision-making, and data-driven system improvement in public health supply chains, especially in low-resource settings. I use data, statistics, and machine learning to address practical supply chain problems where demand is uncertain and operational data are often incomplete. My work combines analytical modelling with practical relevance to support better decisions and meaningful social impact.

Skills & Expertise

Supply Chain	Demand planning; inventory management; supply chain systems
Analytics	Time series forecasting; uncertainty quantification; machine learning; simulation; causal inference; stochastic modelling
Delivery	Stakeholder engagement, knowledge transfer, requirements elicitation, technical documentation, data visualisation
Tools and software	R, Python; Git; Docker; reproducible analytics (Quarto); ML workflows; dashboards (Shiny, Power BI)
Design	Microsoft PowerPoint, Adobe Illustrator, CorelDRAW
Languages	English (Fluent), Sinhala (Native)

Education

PhD in Forecasting & Supply Chain Analytics

Cardiff, UK

Cardiff University, UK

Sep 2023 – Jun 2026 (Expected)

- WGSSS-ESRC Studentship recipient.
- Research: forecasting and decision support to improve reproductive health and family planning supply chains (with USAID) across Côte d'Ivoire, Ethiopia and Laos, working with routine LMIS data and practitioner inputs.

MSc in Social Science Research Methods (Distinction)

Cardiff, UK

Cardiff University, UK

Sep 2022 – Sep 2023

- Dissertation: long-term forecasting and simulation model for NHS workforce planning (with NHS Wales).

MSc in Supply Chain and Data Science (Distinction)

Moratuwa, Sri Lanka

University of Moratuwa, Sri Lanka

Sep 2020 – Jul 2022

- Dissertation: machine learning framework for forecasting retail sales across the promotion life cycle.

BSc (Hons) in Transport and Logistics Management (First Class)

Moratuwa, Sri Lanka

University of Moratuwa, Sri Lanka

Dec 2015 – Jan 2020

- Dissertation: Post-Promotional Effects and Judgmental Adjustments.

Projects & Collaborations

Lead (PhD Project 3)

Data Lab for Social Good

Foundational Time-Series/ LLM Models for Public Health Supply Chains (ongoing)

2025 – Present

- Empirical evaluation using public health supply chain datasets across Ethiopia, Côte d'Ivoire, Laos, and Pakistan; benchmarking accuracy, robustness, and operational feasibility.
- Working paper / ongoing PhD chapter.

Collaboration with HISP Centre, University of Oslo

Oslo, Norway (with country teams)

DHIS2/CHAP Portability for Supply Chain Forecasting in Health Systems

2025

- Evaluated portability of forecasting workflows across morbidity, climate-sensitive diseases, and commodity demand to inform scalable DHIS2/CHAP modules for health system planning.
- Worked closely with the **DHIS2 implementation team, supply chain team, HISP country representatives**, and the **HISP Oslo climate team** to translate stakeholder requirements into methodological choices and system design considerations.
- Translated stakeholder requirements into methodological and system design considerations for national health information systems.
- Manuscript under review at *International Journal of Forecasting*.

Lead (PhD Project 2)

Data Lab for Social Good

Censored Demand Estimation for Contraceptive Supply Chains

2024 – 2025

- Developing methods to estimate true demand under stockouts and interruptions and quantify uncertainty for planning and inventory decisions.
- Designed an operationally deployable workflow using routine LMIS data, focusing on robustness under data censoring and demand volatility.
- Collaborated with **USAID Bureau for Global Health** and the **Côte d'Ivoire country team** to align requirements and validate outputs
- First revision round at *Production and Operations Management (POM)*.

Lead (PhD Project 1)

Data Lab for Social Good

Probabilistic Contraceptive Demand Forecasting

2023 – 2024

- Developed a hybrid forecasting approach combining point forecasts with probabilistic distributions to support contraceptive supply planning under uncertainty.
- Built decision-support outputs to inform stock availability and planning in low-resource settings using routine LMIS data.
- Worked with **USAID Bureau for Global Health** and the **Côte d'Ivoire country team** to align analytical outputs with planning needs and operational constraints.
- Published in *International Journal of Production Research (IJPR)*: <https://doi.org/10.1080/00207543.2025.2526169>.

Collaboration with Ethiopian Pharmaceutical Supply Service (EPSS)

Data Lab for Social Good

Pharmaceutical Supply Chain Forecasting

2024 – 2025

- Co-developed forecasting and decision-support analysis for public healthcare supply chains in Ethiopia, using routine consumption/LMIS data and stakeholder inputs.
- Worked with **Addis Ababa University**, **Ethiopian Pharmaceutical Supply Service (EPSS)**, and the **Ministry of Health** to align methods with operational workflows, data quality constraints, and planning cycles.
- Supported knowledge transfer through technical documentation and hands-on training sessions with local teams.
- Published in *International Journal of Production Research (IJPR)*: <https://doi.org/10.1080/00207543.2025.2546028>.

Collaboration with NHS Wales

Data Lab for Social Good

Long-term Workforce Forecasting & Simulation for Mental Healthcare Planning

2023 – 2025

- Built forecasting and simulation models to support long-term mental healthcare workforce planning and scenario evaluation.
- Linked projections to planning insights for service delivery and capacity decisions.
- Published in *Journal of the Operational Research Society (JORS)*: <https://doi.org/10.1080/01605682.2025.2573825>.

Research outputs (University of Moratuwa; international collaborations)

Retail demand analytics

Retail Demand Forecasting Research (selected)

2021 – 2024

- Studied promotional demand dynamics and forecast adjustments in retail settings, using transactional demand data and time-series forecasting evaluation.
- Collaborations: **Ghent University (Belgium)**; **University of Melbourne (Australia)**.
- Published in *European Journal of Operational Research (EJOR)*: <https://doi.org/10.1016/j.ejor.2021.07.057> and *Journal of Forecasting*: <https://doi.org/10.1002/for.70039>.

Full publication list: [Google Scholar](#).

Work Experience

Research Intern / Visiting Researcher (DHIS2/CHAP)

Oslo, Norway

HISP Centre, Dept. of Informatics, University of Oslo

Sep. 2025 – Dec. 2025

- Worked with **Climate Health Analytics Platform (CHAP)** research and engineering teams to prototype forecasting and inventory optimisation workflows for national health information systems (DHIS2).
- Led a model portability study across morbidity and commodity demand to inform scalable forecasting module design.
- Engaged HISP country teams to capture operational constraints in LMIC supply chains and translate them into implementable system and method requirements.

Senior Business Analyst & Strategist

Colombo, Sri Lanka

Bridge Advisory & Consulting

Jun. 2021 – Dec. 2021

- Led digital transformation initiatives across agriculture and financial services, including e-payments for transportation and a centralised e-KYC platform.
- Supported strategy and delivery for new digital products, translating stakeholder needs into requirements and implementation plans.
- Managed tender documentation processes, ensuring compliance and competitive positioning.

Business Analyst

Colombo, Sri Lanka

ITX360 (Pvt) Ltd

Jun. 2020 – Jun. 2021

- Led a team of business analysts delivering digital solutions across retail, warehousing, healthcare, and e-commerce.
- Managed end-to-end delivery: discovery, functional specs, client workshops, solution pitches, and stakeholder communication.
- Secured and delivered major projects including an agriculture supply chain marketplace and an enterprise system for a national sports organisation.

- Performed inventory audits by reconciling SAGE ERP records with physical counts; identified discrepancies and improved data reliability.
- Built a Power BI KPI dashboard to support supply chain performance monitoring and decision-making.
- Contributed to process review and improvement recommendations for a retail client and a government petroleum institution plan.

Knowledge Transfer & Training

Workshop Facilitator: DHIS2 Workshop on Spatiotemporal Modelling of Climate-sensitive Disease)

Kigali, Rwanda

HISP Centre (University of Oslo) / DHIS2-CHAP Community

Feb. 2026

- Delivered a hands-on session on quantifying uncertainty and tail risks in climate-sensitive disease forecasting, covering analytical distributions, prediction intervals, bootstrap sample paths, and conformal prediction in R.
- Facilitated practical exercises for **60+ participants from 15+ countries**, guiding an end-to-end workflow (model-based intervals → residual bootstrapping → conformal calibration) and addressing implementation questions from practitioners.
- Co-organised with **Rwanda Ministry of Health, HISP Rwanda, CSID Network, Data Lab for Social Good (Cardiff University)**, and **TRUST – Norwegian Centre for Trustworthy AI**.
- [Materials](#)

Workshop Facilitator: Leveraging CHAP and DHIS2 for Stock Forecasting (hands-on)

Online

HISP Centre / DHIS2-CHAP Community

Jan. 2026

- Designed and delivered a practical demo on using the DHIS2 platform and CHAP to generate and interpret stock forecasts.
- Guided participants through an end-to-end workflow (data setup → forecasting → interpretation) and addressed implementation questions from practitioners.
- [Materials](#)

Training Lead: Demand Forecasting Models for Contraceptive Supply Chains (2-day, in-person)

Ethiopia

Ethiopian Pharmaceutical Supply Service (EPSS) & Ministry Stakeholders

Mar. 2025

- Delivered a two-day in-person training on time series forecasting for contraceptive supply chain planning teams.
- Covered model selection and interpretation for operational constraints (demand volatility, stockouts, and supply interruptions), using practical examples aligned to planning workflows.
- [Materials](#)

Instructor

Cardiff, UK

Data Lab for Social Good, Cardiff Business School

Sep. 2023 – Sep. 2025

- Identified training needs and organised training sessions on reproducible research and applied data science for researchers and students.
- Co-developed and delivered hands-on training in **Python, Markov chains, create websites using Quarto** and **causal inference with ML**, supporting participants through exercises, debugging, and implementation questions.
- [Materials](#)

Graduate Tutor (MSc): Forecasting & Business Analytics

Cardiff, UK

Cardiff Business School

Sep. 2023 – Present

- Designed lab materials and delivered sessions in R; supported students through practical problem-solving and feedback on analytical work.

Lecturer (Contract): Supply Chain, IT, and Project Management Modules

Sri Lanka

University of Moratuwa (Dept. of Transport & Logistics Management)

Dec. 2021 – Sep. 2022

- Delivered undergraduate modules and developed assessments; contributed to a professional short course in Strategic Supply Chain Management.
- Supported curriculum development activities, including module updates and programme inputs.

Selected Awards

Mar 2026	Champion for Student Voice and Partnership , Enriching Student Life Awards 2026, Cardiff Students' Union	Cardiff, UK
Feb 2026	Data Lab for Social Good Research Group Excellence Award 2025 , Cardiff University	Cardiff, UK
Sep 2025	Outstanding PhD Peer Mentor Award , Cardiff Business School PhD Peer Mentoring Scheme	Cardiff, UK
Apr 2025	Winner , NATCOR Stochastic Modelling Course Coding Challenge	Lancaster, UK
Jan 2025	Data Lab for Social Good Research Group Excellence Award 2024 , Cardiff University	Cardiff, UK
Jun 2024	Best Research Student Presentation , ISF 2024: International Symposium on Forecasting	Dijon, France
Jan 2021	Logiwiz Award – Best Graduand in Supply Chain Management , University of Moratuwa	Sri Lanka
Jan 2021	SLPA Award – Best Final Year Research Project , University of Moratuwa	Sri Lanka

Professional Memberships & Credentials

Sep 2025	Associate Fellow (AFHEA) , Advance Higher Education Academy	UK
Active	Student Member , The Operational Research Society	UK
Active	Student Member , International Institute of Forecasters	International
Active	Member (MILT) , Chartered Institute of Logistics and Transport	Sri Lanka

Leadership & Service

Programme Manager	Cardiff, UK
Cardiff University Summer School: The Art and Science of Uncertainty and the Future	2025 – Present
- Coordinating programme delivery across partners, speakers, and operational teams; managing timelines, communications, and participant-facing materials. - Supporting outreach and stakeholder engagement to ensure high-quality delivery and accessibility for PhD students and practitioners.	
Research Network Lead	Cardiff, UK
Forecasting for Social Good (F4SG), International Institute of Forecasters	Dec 2021 – Present
- Organized free workshop programme on forecasting in R/Python by recruiting speakers and coordinating delivery (10+ sessions). - Led communications (LinkedIn content, flyers, email coordination) to grow engagement and participation.	
Student Volunteer – Organising Team	Dijon, France
International Symposium on Forecasting (ISF 2024), International Institute of Forecasters	2024
Supported on-site conference operations, including registration and participant coordination for 450+ attendees.	
President	Moratuwa, Sri Lanka
Gavel Club (Toastmasters International), University of Moratuwa	Feb 2019 – Jan 2020
- Led club operations and member engagement; coordinated events, sponsorships, and finances. - Revised the club constitution to improve governance and increase leadership opportunities for student members.	

References upon request